

Lecture 15 Triple Integrals in Spherical Coordinates

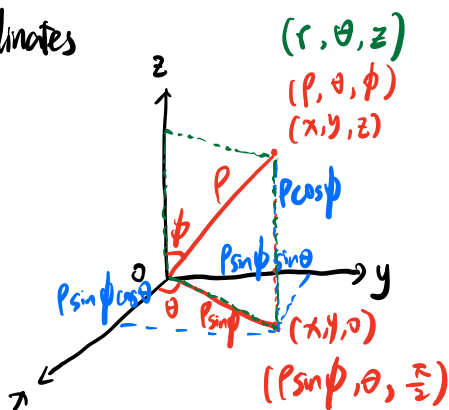
Homework arrangement.

Assignment	Released	Due
6	Oct. 28 (Tuesday)	Nov. 6 (Thursday)
7	Oct. 30 (Thursday)	Nov. 11 (Tuesday)
8	Nov. 6 (Thursday)	Nov. 13 (Thursday)

Recap: In cylindrical coordinates,

$$\iiint_D f(x, y, z) dx dy dz = \iiint_D f(r \cos \theta, r \sin \theta, z) r dr d\theta dz$$

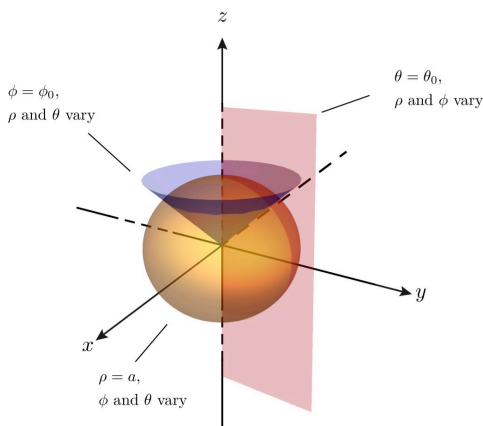
Spherical coordinates



$$\rho \geq 0, 0 \leq \theta \leq 2\pi, 0 \leq \phi \leq \pi$$

$$\begin{cases} x = \rho \sin \phi \cos \theta \\ y = \rho \sin \phi \sin \theta \\ z = \rho \cos \phi \end{cases}$$

$$\begin{cases} r = \rho \sin \phi \\ \theta = \theta \\ z = \rho \cos \phi \end{cases}$$

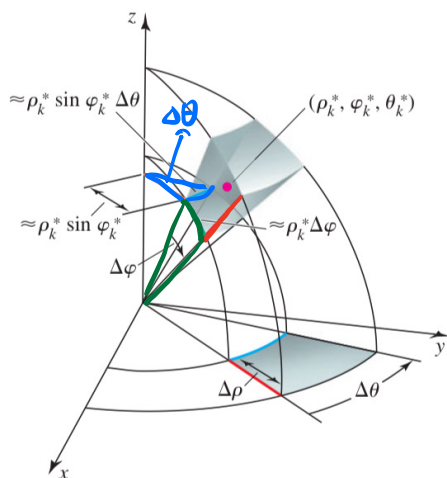


$\rho = \text{constant}$ sphere

$\theta = \text{constant}$ plane

$\phi = \text{constant}$ cone

Theorem. $dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$.



Approximate volume =
 $\Delta V_k \approx \rho_k^{*2} \sin \phi_k^* \Delta \rho \Delta \phi \Delta \theta$

ρ -axis: $\Delta \rho$

ϕ -axis: $\rho \Delta \phi$

θ -axis: $\rho \sin \phi \Delta \theta$

$$\Delta V \approx \rho^2 \sin \phi \Delta \rho \Delta \phi \Delta \theta$$

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

Jacobian matrix $x(u, v, w) \quad y(u, v, w) \quad z(u, v, w)$

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{pmatrix}$$

$$dx \, dy \, dz = \left| \det \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du \, dv \, dw$$

$$\frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} = \begin{pmatrix} \frac{\partial x}{\partial \rho} & \frac{\partial x}{\partial \phi} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial \rho} & \frac{\partial y}{\partial \phi} & \frac{\partial y}{\partial \theta} \\ \frac{\partial z}{\partial \rho} & \frac{\partial z}{\partial \phi} & \frac{\partial z}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \sin \phi \cos \theta & \rho \cos \phi \cos \theta & -\rho \sin \phi \sin \theta \\ \sin \phi \sin \theta & \rho \cos \phi \sin \theta & \rho \sin \phi \cos \theta \\ \cos \phi & -\rho \sin \phi & 0 \end{pmatrix}$$

$$\begin{aligned} \det \frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} &= \sin \phi \cos \theta \det \begin{pmatrix} \rho \cos \phi \sin \theta & \rho \sin \phi \cos \theta \\ -\rho \sin \phi & 0 \end{pmatrix} \\ &\quad - \rho \cos \phi \cos \theta \det \begin{pmatrix} \sin \phi \sin \theta & \rho \sin \phi \cos \theta \\ \cos \phi & 0 \end{pmatrix} \\ &\quad - \rho \sin \phi \sin \theta \det \begin{pmatrix} \sin \phi \sin \theta & \rho \cos \phi \sin \theta \\ \cos \phi & -\rho \sin \phi \end{pmatrix} \end{aligned}$$

$$\begin{aligned}
&= \rho^2 \cos^4 \phi \sin \phi \cos^2 \theta + \rho^2 \sin^3 \phi \sin^2 \theta \\
&\quad - (-\rho^2 \cos^2 \phi \sin \phi \sin^2 \theta - \rho^2 \sin^3 \phi \cos^2 \theta) \\
&= \rho^2 \cos^2 \phi \sin \phi + \rho^2 \sin^3 \phi \\
&= \rho^2 \sin \phi.
\end{aligned}$$

Example. Consider a solid sphere D of radius r centered at the origin. Suppose it has uniform density δ . Evaluate

$$I_z := \iiint_D \delta(x^2 + y^2) dv. \quad \text{moment of inertia.}$$

Sol. $0 \leq \rho \leq r, \quad 0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq \pi$

$$x^2 + y^2 = (\rho \sin \phi \cos \theta)^2 + (\rho \sin \phi \sin \theta)^2 = \rho^2 \sin^2 \phi$$

$$I_z = \delta \int_0^{2\pi} \int_0^\pi \int_0^r \rho^2 \sin^2 \phi \cdot \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

$$= \delta \left(\int_0^{2\pi} d\theta \right) \cdot \left(\int_0^\pi \sin^3 \phi \, d\phi \right) \left(\int_0^r \rho^4 \, d\rho \right)$$

$$= \delta \cdot 2\pi \left(-\int_0^\pi (1 - \cos^2 \phi) d\cos \phi \right) \frac{r^5}{5}$$

$$= \delta \cdot 2\pi \left[-\cos \phi + \frac{\cos^3 \phi}{3} \right]_0^\pi \cdot \frac{r^5}{5}$$

$$= \delta \cdot 2\pi \left(1 - \frac{1}{3} + 1 - \frac{1}{3} \right) \cdot \frac{r^5}{5}$$

$$= \frac{8\delta\pi r^5}{15}$$

$$\delta = \frac{m}{V}$$

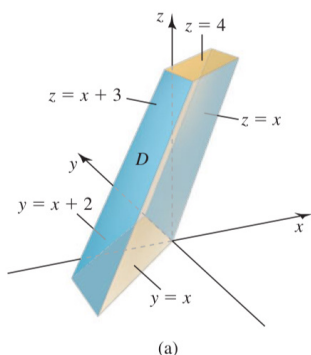
$$I_z = \frac{8\pi r^5}{15} \cdot \frac{\frac{3}{4}\pi r^3}{4\pi r^3} = \frac{2mr^2}{5}.$$

$$V = \iiint_D 1 \, dv = \int_0^{2\pi} \int_0^\pi \int_0^r \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

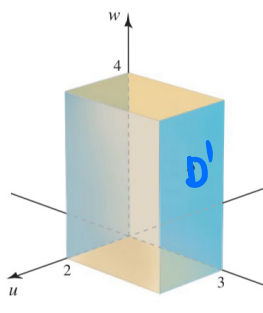
$$= \int_0^{2\pi} d\theta \int_0^\pi \sin \phi \, d\phi \int_0^r \rho^2 \, d\rho = 2\pi \cdot 2 \cdot \frac{r^3}{3} = \frac{4\pi r^3}{3}.$$

Example. Evaluate $\iiint_D xz \, dV$, where D is a parallelepiped bounded by the planes

$$y=x \quad y=x+2 \quad z=x \quad z=x+3 \quad z=0 \quad z=4$$



(a)



(b)

Sol. $y-x=0$ & $y-x=2$

$$z-x=0 \quad \& \quad z-x=3$$

$$z=0 \quad \& \quad z=4$$

$$u=y-x, \quad v=z-x, \quad w=z$$

$$D' = \{(u, v, w) : 0 \leq u \leq 2, 0 \leq v \leq 3, 0 \leq w \leq 4\}$$

$$\begin{cases} x = w - v \\ y = u - v + w \\ z = w \end{cases}$$

$$\det \frac{\partial(x, y, z)}{\partial(u, v, w)} = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{pmatrix}$$

$$= \det \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = 1$$

$$xz = (w-v)w = w^2 - vw$$

$$\iiint_D xz \, dV = \iiint_{D'} (w^2 - vw) \, du \, dv \, dw$$

$$= \int_0^4 \int_0^3 \int_0^2 (w^2 - vw) \, du \, dv \, dw$$

$$= 2 \int_0^4 \left[w^2 v - \frac{1}{2} v^2 w \right] \Big|_0^3 \, dw$$

$$= 2 \int_0^4 \left(3w^2 - \frac{9}{2} w \right) \, dw$$

$$= 2 \left[w^3 - \frac{9}{4} w^2 \right] \Big|_0^4 = 56$$